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## **Diagenetic Control on Rock-Type Heterogeneity in Dolomitized Upper Triassic Jilh Formation (Saudi Arabia) Revealed by *In-Situ* LA-ICP-MS U/Pb Dating**

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### **Abstract**

The Upper Jilh Formation was deposited during the Late Triassic (~238Ma - ~228Ma) within a broad peritidal to shallow marine carbonate platform. Facies are pervasively dolomitized and locally associated with good development of intercrystalline macropores, mainly in initially grainy facies. This study aims to provide new geochronology insights into multiple dolomitization processes affecting the Upper Jilh Formation and to establish connections between diagenetic overprinting and large-scale rock-type variability.

Core and petrographic observations of 580 thin-sections from 12 wells enable the reconstruction of the paragenetic sequence. Stable isotopes ( $\delta^{18}\text{O}$  &  $\delta^{13}\text{C}$ ) were carried out on more than 850 samples and coupled with LA-ICP-MS in-situ U/Pb chronometry on 10 samples, along with one sample analyzed for fluid inclusion microthermometry. This compelling geochemical dataset was also integrated with plug measurements, including 53 MICP points, to define core-based rock types (RRTs) characterized by distinct petrographic, chronographic, diagenetic, and pore system attributes.

Analyses indicate at least three phases of dolomites. Planar-s to non-planar dolomites (rD1) are usually medium crystalline and  $\delta^{18}\text{O}$  composition between -4‰ and 0‰. U/Pb chronometry indicates a precipitation within a narrow timeframe during the Late Triassic ( $214 \pm 7.6$  Ma,  $n = 7$ ). rD1 is usually associated mud-dominated fabrics and show limited small intercrystalline macropores. Porous planar-e dolomites (rD2) are more common within grainstones and display coarser crystals with rare locally ferroan overgrowths. This phase has  $\delta^{18}\text{O}$  values between -6‰ and -1‰, and preserved a LA-ICP-MS age of  $171 \pm 7.8$  Ma ( $n=5$ ) indicating a broad Early Jurassic precipitation. These dolomites are associated with larger intercrystalline macropores with larger pore throat diameters (RRT3). Burial cements include large mm-scale non-ferroan saddle dolomites with very negative oxygen isotopes (average -9.5‰) precipitated by hot brines (~160°C, fluid salinity of 23.2 wt.% NaCl eq.) likely circulating along faults. This phase was not directly datable by U-Pb geochronology. Based on its paragenetic relationship with other phases, its precipitation age is estimated between 171 Ma and 135 Ma. Indeed, the last dated diagenetic event is an Early Cretaceous (~135 ± 20 Ma) blocky calcite associated with pore-filling anhydrite, locally replaced by calcite due to limited TSR.

Rock-typing analysis reveals that pore systems are influenced by both sedimentological and diagenetic factors and can be reliably predicted in uncored sections. For the first time on the Arabian Platform, *in-situ* U/Pb chronometry has been used to date the key pore enhancing and pore occluding diagenetic phases. These findings provide valuable data for integration with basin modeling to improve reservoir characterization and predictive model accuracy.

## Introduction

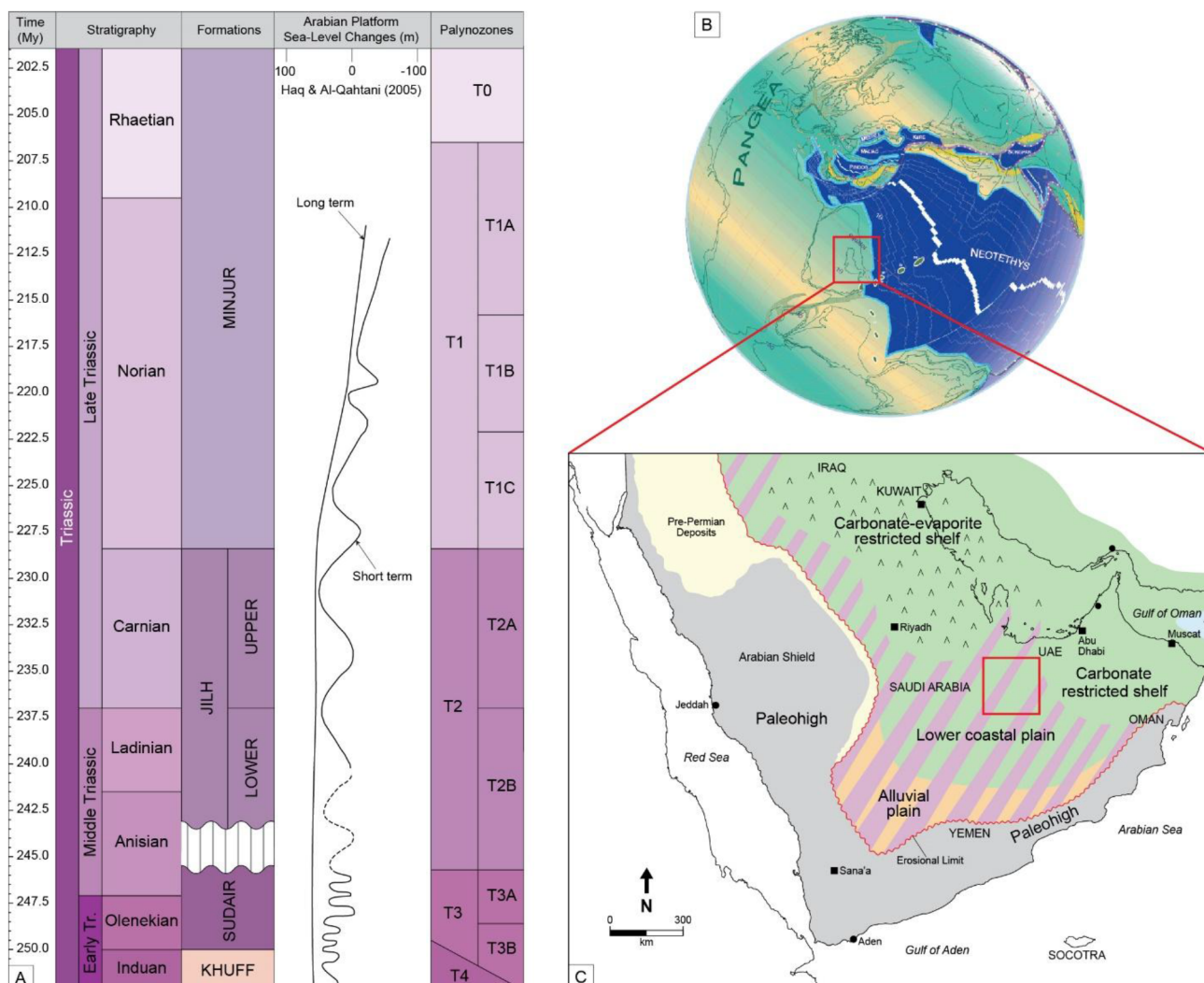
Rock-typing studies that include geological factors such as facies variations and diagenetic overprint usually focus on predicting the distribution of the pore volume across a given stratigraphic interval, but not necessarily on the timing of formation of the porosity, which remains critical, especially for exploration purposes. Amongst carbonate systems, dolomitized deposits typically host fairly "simple and predictable" pore systems, compared to limestones, dominated by intercrystalline macropores and locally associated with other pore types. Whilst dolomitization models and diagenetic interpretations have traditionally integrated petrography, stable isotopes, and fluid inclusion data, they often lack recent methodological advancements in diagenetic studies, such as *in-situ* U/Pb chronometry by Laser Ablation Induced Plasma Mass Spectrometry (LA-ICP-MS). Consequently, many models rely on imprecise relative timing, leaving the possibility for refinement based on more recent methodological advancements. In this context, new geochronology insights into multiple dolomitization processes affecting the Upper Jilh Formation are needed to establish connections between diagenetic overprinting and large-scale rock-type variability.

## Geological background

Detailed geological setting of the Jilh formation is provided in [Deville de Periere et al. \(2022\)](#) and is only summarized here. The Jilh Formation, which spans from Anisian to Carnian (~238Ma - ~228Ma) are deposited on the Arabian Platform (western margin of the Neotethys Ocean - [Stampfli & Borel, 2002](#) – [Figure1B](#)) under warm and arid to possibly humid tropical climate. Average seawater temperatures reached 28°C to 35°C according to regional paleoclimatic models (e.g., [Sellwood & Valdes, 2006](#); [Li et al., 2022](#)).

In the south and south-west of Saudi Arabia, alluvial and coastal plain facies dominated by clastic deposits are present, likely associated with the erosion of the exposed Arabian Shield ([Obermaier et al., 2012](#) – [Figure1C](#)). These deposits transition to dolomitized mudstones and microbial intertidal facies deposited in a large peritidal environment. Large, shallow marine inner ramp deposits are characterized basinward by peloidal and oolitic banks ([Obermaier et al., 2012](#); [Deville de Periere et al., 2022](#)). In the United Arab Emirates, inner ramp facies laterally pass to platform margin clinoforms prograding towards an intrashelf basin identified from seismic data (see [Hu et al., 2015](#); [Taher et al., 2017](#)).

According to [Deville de Periere et al. \(2002\)](#), the Jilh Formation is divided into four 3<sup>rd</sup> order sequences that can be traced across Saudi Arabia (see also [Al-Saqer et al., 2015](#)). The Upper Jilh Formation encompasses sequences 3 and 4 and is dated from the Carnian according to detailed palynological observations (T2A palynozone - [Deville de Periere et al., 2022](#)).



**Figure 1—A. Simplified stratigraphy of the Triassic in Saudi Arabia including eustatic variations of the Arabian Platform (Haq & Al-Qahtani, 2005) and Triassic palynozones (partially published in Issautier et al., 2019); B. Global paleogeographical map focusing on the Neotethys at the top of the Carnian, ~200My (Stampfli & Borel, 2002); (C) Middle Triassic depositional environment distribution of the Arabian plate. Modified from Obermaier et al., 2012 and Ziegler, 2001). Figure from Deville de Periere et al. (2022).**

## Material and methods

The present study is based on the integration of 12 wells, including the detailed sedimentological description of approximately 400m of slabbled cores. Thin sections (n=580) were prepared from core chips and impregnated with blue-dyed epoxy. Wireline log data, including gamma ray and neutron porosity/density were available for all wells. Cathodoluminescence (CL) observations were performed on a subset of 30 polished thin-sections. Core plugs and associated porosity/permeability data were available every 30cm, and 53 Mercury Injection Capillary Pressure samples were also analyzed to assess pore throat diameters and distribution.

Additionally, 780 samples were prepared for bulk stable isotopes ( $\delta^{18}\text{O}$ /  $\delta^{13}\text{C}$ ), targeting homogenous replacive dolomites, while 9 samples were analyzed targeting major cementation phases (calcite and dolomite). Analyses were performed using a Thermo-Fisher Gas Bench coupled with a Thermo-Fisher MAT253 mass spectrometer in Saudi Aramco EXPEC-ARC laboratory, with accuracy of  $\pm 0.1$  ‰ for both  $\delta^{18}\text{O}$  and  $\delta^{13}\text{C}$ . Ten polished thin-sections were used for *in-situ* U/Pb age dating using Laser Ablation Induced Plasma Mass Spectrometry (LA-ICP-MS). This was done at the Centre Européen de Recherche et d'Enseignement des Géosciences de l'Environnement (CEREGE), Aix-en-Provence, France. An NWR

193 nm (Elemental Scientific) link to an Element XR SF-ICP-MS (Sector Field ICP-MS - Thermo Fisher Scientific) was used for these analyses. Fluid inclusions were analyzed on one thick section (150  $\mu\text{m}$ ), through a LINKAM MDS 600 heating-freezing stage, analyzing both melting and freezing temperatures. More details on these methods can be found in [Deville de Periere et al. \(2025\)](#).

## Sedimentology of the Upper Jilh Formation

The studied interval accounts for the base of the Carnian stage (Sequence 3) and can be subdivided into three units based on sedimentological changes and gamma-ray patterns. The detailed description of this sequence was provided in [Deville de Periere et al. \(2022\)](#) and is only summarized hereafter. The lowermost unit is dominated by peritidal dolomitized mudstones interbedded with abundant, m-scale thick, laterally extensive clastic-rich mudrocks and minor nodular anhydrite beds associated with high gamma-ray values. These facies likely represent sabkha and tidal flat environments, possibly reaching a palustrine setting. The top of this unit, marked by a sharp decrease in gamma-ray values is characterized by a grain-dominated bed (possible washovers?) including common peloids and intraclasts, along with rare bivalves, echinoderms and small benthic foraminifera. The middle unit is characterized by a sharp decrease in gamma-ray that can be traced regionally in Syria, Kuwait, UAE and Oman (see [Lučić et al., 2010](#); [Obermaier et al., 2012](#); [Hu et al., 2015](#), [Husain et al., 2018](#)). Overall, facies are dominated by bioturbated mudstones (dolomitized or not) with minor microbial laminations, and peloidal wackestones and packstones. These grainy peloidal deposits also include echinoderm debris, coated-grains and intraclasts, and suggest the development of grainy banks in a low to moderate-energy inner ramp setting, and likely mark the major maximum flooding interval of the entire Jilh Formation. The top of this middle unit includes an approximately 3m-thick limestone marker bed characterized by pervasively bioturbated and wispy laminated partially dolomitized mudstone with echinoderm and ostracod debris. The uppermost interval is marked by an increase in gamma-ray values, associated with the deposition of clastic-rich mudrocks interbedded with dolomitized mudstones. Carbonate beds are dominated by massive, fine-crystalline dolomitized mudstones along with finely laminated mudstones with clastic material likely indicating tidal influence. Glauconite grains are locally present in cm-scale lag beds associated with echinoderm and bivalve debris.

## Diagenesis: Petrography and geochemistry

In the Upper Jilh, petrographical observations, along with geochemistry data show at least 17 diagenetic phases, including dolomitization, calcite and dolomite cementation and dissolution events. The following paragraphs summarize the petrographical and geochemical characteristics of the key diagenetic phases.

### Replacive dolomite

The Upper Jilh is pervasively dolomitized and two types of replacive dolomite fabrics have been described. Planar-s to non-planar dolomites (rD1) are usually medium crystalline and bright to dull red under cathodoluminescence. This replacive dolomite is usually observed within the bioturbated mudstones located in the lowermost part of the studied interval.  $\delta^{18}\text{O}$  and  $\delta^{13}\text{C}$  composition ranges between -4‰ and 0‰ and between +2‰ and +4‰, respectively. U/Pb chronometry indicates a precipitation within a narrow timeframe during the Late Triassic ( $208 \pm 16$  Ma,  $n = 7$ ). This replacive dolomite is interpreted to be formed under limited burial conditions, likely from modified seawater associated with the Triassic transgression of the Minjur Formation. Porous planar-s to planar-e dolomites (rD2) are more common within grain-dominated facies observed at the vicinity of the main maximum flooding event. rD2 is medium to coarse-crystalline with a dull red luminescence. Rare thin-lipid overgrowths with a bright red luminescence are locally observed. This phase has  $\delta^{18}\text{O}$  values between -6‰ and -1‰ and  $\delta^{13}\text{C}$  between +3‰ and +4‰. LA-ICP-MS chronometry indicates a precipitation during the early Jurassic with an age of  $177 \pm 13$  Ma ( $n=5$ ). This phase is a burial dolomitization event and the origin of the magnesium-rich is unclear at this stage. The



age of the two replacive dolomite events has been tested in various anticline structures of the Rub al Khali Basin located tens of kilometers away from each other. U/Pb chronometry shows that the ages are similar across the basin, suggesting regional trends of dolomitization, possibly affecting the entire platform, during both late Triassic and Lower Jurassic.

As rD2 is mainly observed within the grainiest fabric, it is possible that rD1 phase initially dolomitized the entire Upper Jilh section, but grainy deposits that kept open porosity during the burial history likely got preferentially affected by the second dolomitization event under burial condition.

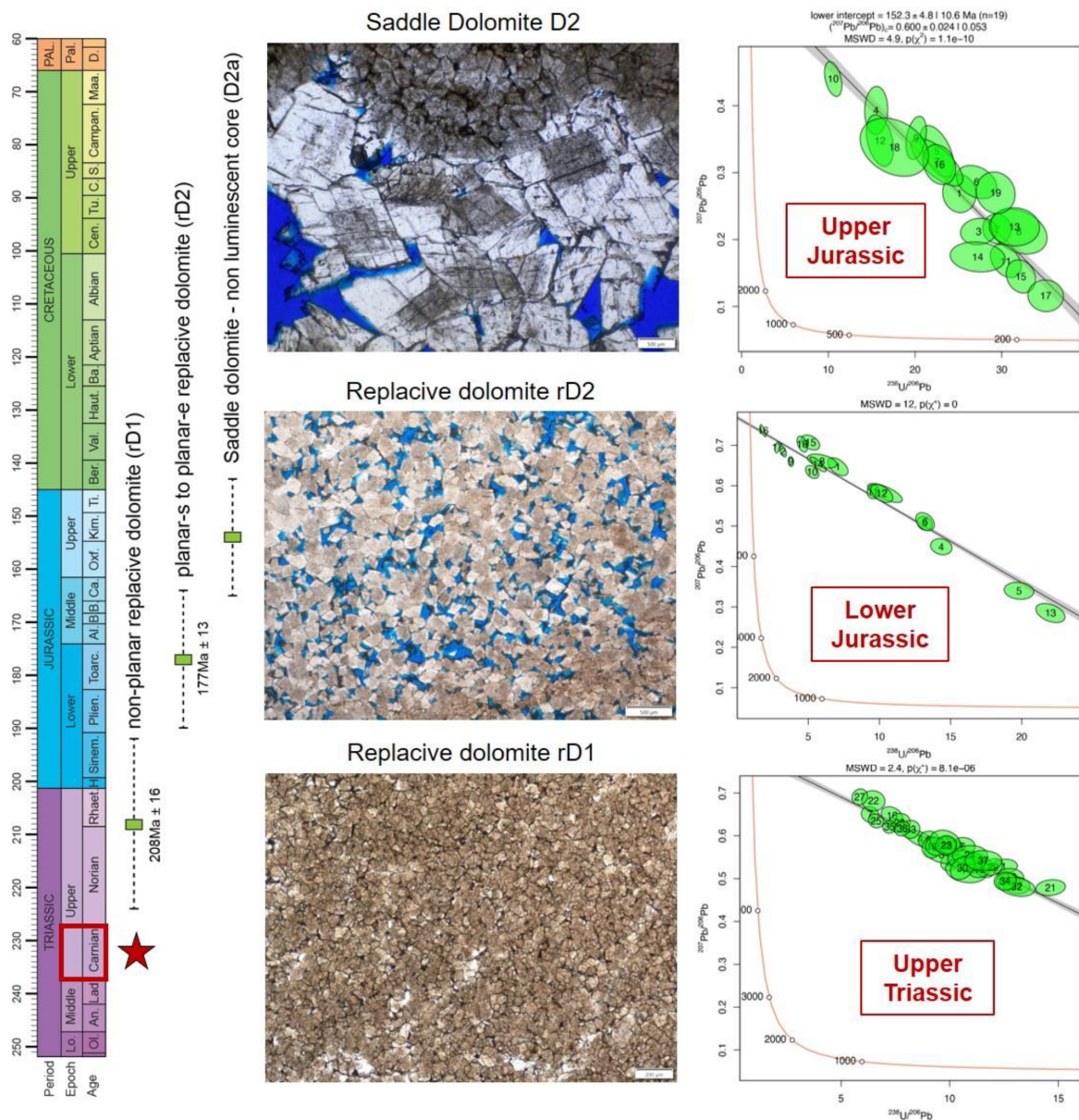


Figure 2—Key dolomitization and dolomite cement phases along with their representative U/Pb results. The red stars indicates the time of deposition inferred by palynology.

Within the Upper Jilh, the limestone marker bed is locally partially dolomitized and show the presence of minor to common euhedral, fine to medium crystalline rhombs, which are usually weakly ferroan. Although these dolomite rhombs could not be dated by LA-ICP-MS, the micritic matrix of the limestone yielded an Upper Triassic age ( $226 \pm 17$  Ma; n=1), which includes the depositional time of the Upper Jilh.  $\delta^{18}\text{O}$  values

in the limestone bed range between -8.5‰ and -2‰, while  $\delta^{13}\text{C}$  varies between +2‰ and +4.5‰. This suggests a very shallow burial recrystallization of the micritic matrix, possibly from meteoric fluids.

### Burial cements

Dolomite cements are characterized by large, locally mm-scale, non-ferroan saddle dolomites (D2). The core of these dolomite is locally inclusion-rich with a dull red to brown luminescence (D2a), but the majority of the rhombs are inclusion-free with a bright red luminescence under CL (D2b). Saddle dolomites are observed as a pore occluding phases, mainly in vugs, but is also present in "zebra-like" structure observed in conjunction with fractures in the lowermost part of the section. Stable isotopes measured on D2 show negative  $\delta^{18}\text{O}$  (average -9.5‰) and an  $\delta^{13}\text{C}$  of +2.5‰. fluid inclusions indicate the D2 precipitated by hot brines (~160°C, fluid salinity of 23.2 wt.% NaCl eq.) likely circulating along faults. These dolomites are considered hydrothermal as their precipitation temperature is ~60°C warmer than the host rock according to basin models. U/Pb chronometry in D2a (n=1) indicates an Upper Jurassic age of precipitation ( $154 \pm 11$  Ma), but D2b could not be directly dated by U-Pb geochronology. Based on its paragenetic relationship with other phases, its precipitation age is estimated between 154 Ma and 135 Ma (Upper Jurassic / Early Cretaceous). Indeed, the last dated diagenetic event is an Early Cretaceous (~135 ± 20 Ma) blocky calcite (C4) associated with pore-filling anhydrite. Under CL, C4 shows a dull orange to brown luminescence, and the only sample measured for stable isotopes indicates a very negative  $\delta^{18}\text{O}$  of -11.5‰ and a  $\delta^{13}\text{C}$  of -0.5‰. Anhydrite cements are rare overall, and large crystals are locally replaced by calcite and native sulfur due to limited TSR.

## Rock-typing: depositional and diagenetic controls

### rock-typing of dolomitized fabrics: depositional and diagenetic controls

The rock-typing presented in the present study is based on this integration of core and thin-section observations, along with plug data such as porosity, permeability and MICP data. Rock-types (RRTs) should honor log data but must have a geological meaning dictated by the rock observations. In the studied interval, three main rock-types have been described within the dolomitized units. It has to be noted that as the limestone marker bed is always pervasively cemented and tight, it has been removed from the below rock-typing study, which in turn only focuses on dolomitized fabrics. The identified rock-types can be summarized as follows:

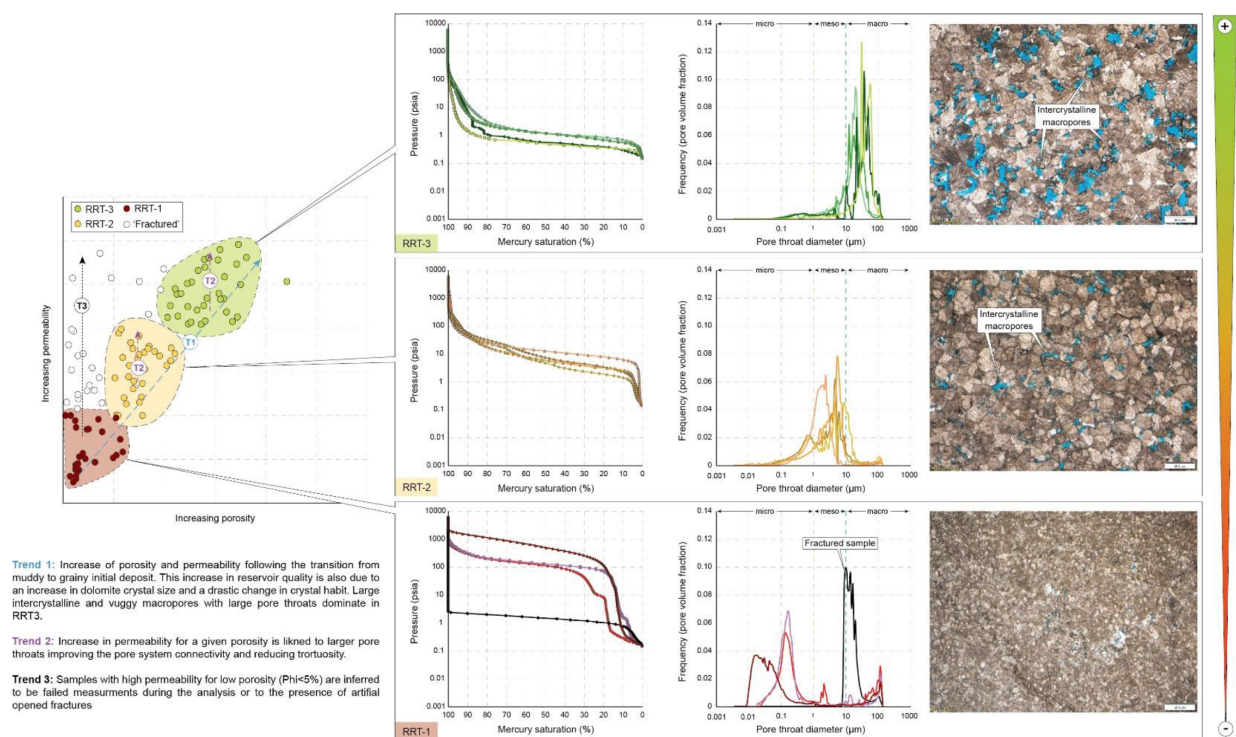
- RRT-1 encompasses tight dolomitized mudstones and wackestones deposited in a broad tidal flat environment. Dolomitized fabrics are usually non-planar to planar-s and coincide with dolomitization event rD1. Dolomite rhombs are fine crystalline (usually <100µm). Macropores are generally pervasively occluded by saddle dolomite (D2) and anhydrite and more rarely by blocky calcite. The visible pore system includes rare small intercrystalline macropores and micropores. MICP data show that the average pore throat diameter is <0.5µm.
- RRT-2 is characterized by dolomitized mudstones to peloidal packstones mainly deposited in tidal flat to possibly low-energy inner ramp setting. The dolomite fabric is typically planar-s, with medium-crystalline rhombs usually <150µm. This rock type is usually dominated by rD2 dolomitization event. Minor to locally common intercrystalline macropores account for the visible pore volume, with an average pore throat diameter of ~6µm. The largest macropores are usually cemented by dolomite and anhydrite. The pore throat distribution shows a wide range of diameters, typically between 0.5µm and 10µm, likely creating a heterogeneous pore system at the plug-scale.
- RRT-3 is dominated by dolomitized peloidal packstone and grainstones deposited in low to moderate-energy inner ramp domain. The replacive dolomite fabric is typically planar-e with subordinate planar-s and dolomite rhombs are medium to coarse-crystalline (<150µm). This rock-

type is directly related to rD2 dolomitization. Common to abundant intercrystalline macropores, along with subordinate vugs, account for the visible pore volume. Cementation of macropores by saddle dolomite is very limited. Pore throat diameters are usually large, with an average diameter greater than  $\sim 80\mu\text{m}$ , while their distribution is narrower than that of RRT-2, suggesting a more homogeneously distributed pore system.

These rock types are primarily a function of the original depositional fabric, with RRT-1 and RRT-2 being dominated by mud-dominated facies and RRT-3 associated with grainier textures. Dolomitization phases, however, played a critical role in improving or at least preserving the pore system. Hence, the best rock-type (RRT-3) appears related to the burial dolomitization event that took place during the Lower/Middle Jurassic, while the poorer quality rock-types appears related to earlier dolomitization processes. This is the first time on the Arabian Platform that the origin of rock-types is tightly calibrated to the geological history of the studied basin.

### Prediction of rock-types: from core to log-scale

To be useful in subsurface studies, rock-types need to be predictable in uncored section, primarily using log data. Defining log-based rock-type (i.e., electrofacies) is always extremely challenging in carbonate successions, mainly due to the limited changes in mineralogy and the large variation in pore systems within given unit. Therefore, rock-typing studies and predictability of the defined rock-types are usually complex, time consuming and likely to fail. In the Upper Jilh, however, the pore systems and facies are rather "simple" and when compared to basic log data such as neutron and density (NPHI/RHOB), it is possible to easily define cutoff values that can be used to predict each rock-type at the well-scale and beyond.



## Conclusions

The Upper Jilh Formation (Upper Triassic) can be divided into three units, each with distinct facies and gamma-ray patterns. The lowermost unit is characterized by peritidal dolomitized mudstones, while the



middle unit is marked by a sharp decrease in gamma-ray values and the development of grainy banks in a low to moderate-energy inner ramp setting. The uppermost interval is marked by an increase in gamma-ray values and the deposition of clastic-rich mudrocks. Diagenetic events, mainly including dolomitization and cementation, have played a critical role in the distribution of the pore volume. LA-ICP-MS in-situ U/Pb chronometry, along with stable isotopes, petrography and fluid inclusions, support the identification and time constraint of two replacive dolomite phases and several cementation events including hydrothermal saddle dolomite and calcite cements. Three main rock-types have been identified, each with distinct depositional and diagenetic controls. The best rock-type is related to the burial dolomitization event that took place during the Lower/Middle Jurassic. This study provides new insights into the geological history of the Upper Jilh Formation and has implications for subsurface studies. The agedating of the dolomitization events and the predictability of rock-types using log data are significant outcomes, allowing for more accurate characterization of the formation at the platform-scale.

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